

Jack Brett

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Education

Bournemouth University

Engineering Doctorate alongside [ROLI](#)

Poole, UK

2019 – 2023

Bournemouth University

Games Technology (First Class Honours BSc)

Poole, UK

2014 – 2017

Experience

Postdoctoral Research Assistant in Authoring Tools for Locative Games

August 2023 – Present

Bournemouth University - LoGaCulture

Poole, UK

- Spearheading [authoring tool development](#) in Unity for [LoGaCulture](#), an EU-funded initiative, collaborating with a large number of organisations and universities; released several games on [iOS](#) and [Android](#)
- Crafting immersive mixed reality experiences for cultural heritage, working closely with the National Trust
- Lead developer and researcher role (Photoshop, git, Unity (plug-in development), C#, JavaScript, TypeScript, React, SQL, and C++)

Research Engineer and Doctoral Student

October 2019 – December 2023

ROLI & Bournemouth University

London, UK

- Worked with [ROLI](#) during an EngD placement supported by the [Centre for Digital Entertainment](#)
- Conducted research on the intersection of music education and video games, emphasising rote learning
- Utilised multiple game development tools to create training games, leading four user studies resulting in a series of [published papers](#) including conferences and journals
- Collaborated with the R&D team and lead developers at ROLI (C#, C++, TypeScript, C, Unity, UE4, Blender)

Lead Research Developer

December 2017 – July 2022

Vitae VR & King's College London

Bournemouth, UK

- Collaborated with VitaeVR, a subsidiary of [Outpost VFX](#), and [KCL Psychology Department](#)
- Facilitated the transfer of [VStore](#) from PC to PSVR as well as leading development of key aspects of the program using Blender, Maya, UE4 and C++ with visual scripting; utilised Javascript, Python for web and API.
- [VStore](#) is a novel functional cognition VR shopping task created in UE4 with the intent to increase the ecological validity of cognitive assessments, specifically Dementia and Alzheimer's
- Served as an independent consultant and developer, collaborating with the [lead researcher](#) and the team at [King's College University and Hospital](#)

Projects

Scale Loop | C#, shaderLABS, Unity, C++, Python, Photoshop, Git

2021 – Present

- Developed a synthesiser application using Unity where you can play in different musical scales from cultures around the world and record loops using different sounds
- Inspired by my experience at ROLI and a fascination of scales, [Looper](#) attempts to teach people about musical scales in a whimsical way
- Currently released on the [Microsoft Store](#) and [Google Play Store](#).

Bridge Wavers | C#, JavaScript, shaderLABS, Git, Photoshop, Procreate

2020 – Present

- Another game under the name of ['Boggy Entertainment'](#)
- [The game](#) is all about navigating an increasingly dangerous bridge to wave at cars below. Inspired by pandemic strolls with the family where we would actually wave at cars from a local bridge!
- The game was developed using scriptable object architectures to expose runtime events in a simple way and to avoid having to delve into code to make objects communicate with each other (helping to avoid overusing the singleton pattern a lot of Unity games seem to rely upon)
- Releases coming soon!

Publications

[LUTE: A Hypertextual Mixed Reality Game Engine](#) (ACM Hypertext '25)

[Understanding Attitudes on Mixed Reality Heritage](#) (NHT'25 Workshop at ACM Hypertext 2025)

[Authoring Tools for Mixed Reality](#) (NHT'23 Workshop at ACM Hypertext 2023)

[Graveyard Symphony; A Study Investigating the Effectiveness of a Prototype Music Game](#) (Games: Research and Practice)